Akoizo Original Practice - 2027 Rotation Form 2

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A station model pressure code of 984 means:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. 1008.4 mb

B. 998.4 mb

C. 1098.4 mb

D. 984 mb

Answer: _____

2. (2 points) Which air mass is warm and humid?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Continental tropical

B. Continental arctic

C. Maritime tropical

D. Continental polar

Answer: _____

3. (2 points) Damage spreading outward in a starburst pattern indicates:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. A tornado

B. Hail

C. A dryline

D. A downburst

Answer: _____

4. (2 points) Sleet forms when:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Melted snow refreezes in a deep cold layer near the ground

B. Snow never melts

C. Rain freezes on contact

D. Hail falls from a supercell

Answer: _____

5. (2 points) In a METAR, “31/22” means:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. Temperature 31 °C, dew point 22 °C

B. Wind 310° at 22 knots

C. Visibility 31 miles

D. Pressure 31.22 inches

Answer: _____

6. (2 points) A dryline separates air masses that differ mainly in:

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Wind speed only

B. Moisture

C. Temperature

D. Pressure only

Answer: _____

7. (2 points) The EF scale estimates wind speed from:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. Radar reflectivity

B. Funnel width

C. Observed damage to indicators

D. Direct anemometer readings in the tornado

Answer: _____

8. (2 points) A tornado warning means:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Conditions may become favorable later

B. The storm has ended

C. Only hail is expected

D. A tornado is occurring or imminent; take shelter now

Answer: _____

9. (2 points) A long hodograph that curves clockwise near the ground indicates:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. No wind

B. Stable air

C. Calm conditions favoring fog

D. Strong shear favoring supercells

Answer: _____

10. (2 points) Why do hurricanes weaken after landfall?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
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Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. They lose their warm-water energy source

B. Land air is too moist

C. The jet stream stops them

D. Coriolis force becomes zero

Answer: _____

11. (2 points) Interpret the adjacent opposite radial velocities cautiously.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

13. (2 points) Calculate the surface temperature-dew-point spread.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

15. (2 points) A storm moves 30 km east in 30 minutes. Find mean speed.

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 30 km

Answer: _____
_____
_____

16. (2 points) A rain gauge records 40 mm in 0.5 hours. Calculate mean rain rate.

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 40 mm

Answer: _____
_____
_____

17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

Answer: _____
_____
_____

18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

Answer: _____
_____
_____

19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

Answer: _____
_____
_____

20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

Answer: _____
_____
_____